SOC Lab – Attack Detection Report

Executive Summary

This report documents simulated attacks performed from Kali Linux against Ubuntu and Windows machines. The objective was to evaluate Wazuh's ability to detect brute-force and reconnaissance activities. Multiple attacks including SSH brute force, RDP brute force, and network enumeration were executed. Wazuh successfully detected authentication failures and suspicious login attempts. Alerts were analyzed and categorized, and mitigation recommendations were proposed.

Scope

The lab environment consisted of three virtual machines:

- Kali Linux (Attacker)
- Ubuntu (Target)
- Windows 11 (Target)
- Wazuh Server (Monitoring)

Environment Setup

Network Configuration:

- Kali Linux (Attacker & Wazuh Server): 192.168.56.103
- Ubuntu Target: 192.168.56.110
- Windows Target: 192.168.56.105
- Metasploitable: 192.168.56.101

Wazuh agents were installed on Ubuntu and Windows systems.

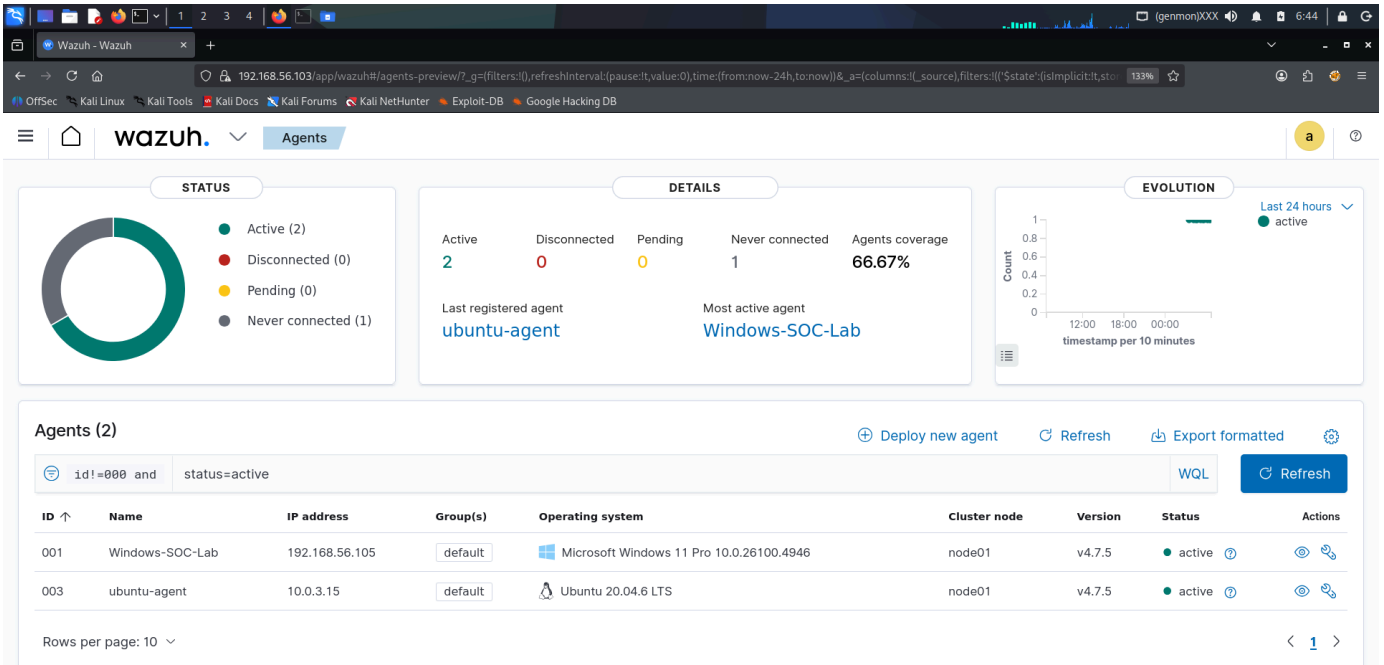

Figure 1: Wazuh agents connected to manager

Attack Simulation

1. Network Enumeration

An Nmap scan was performed to identify open ports and services.

Timestamp: 06:45

Command: sudo nmap -p- 192.168.56.101

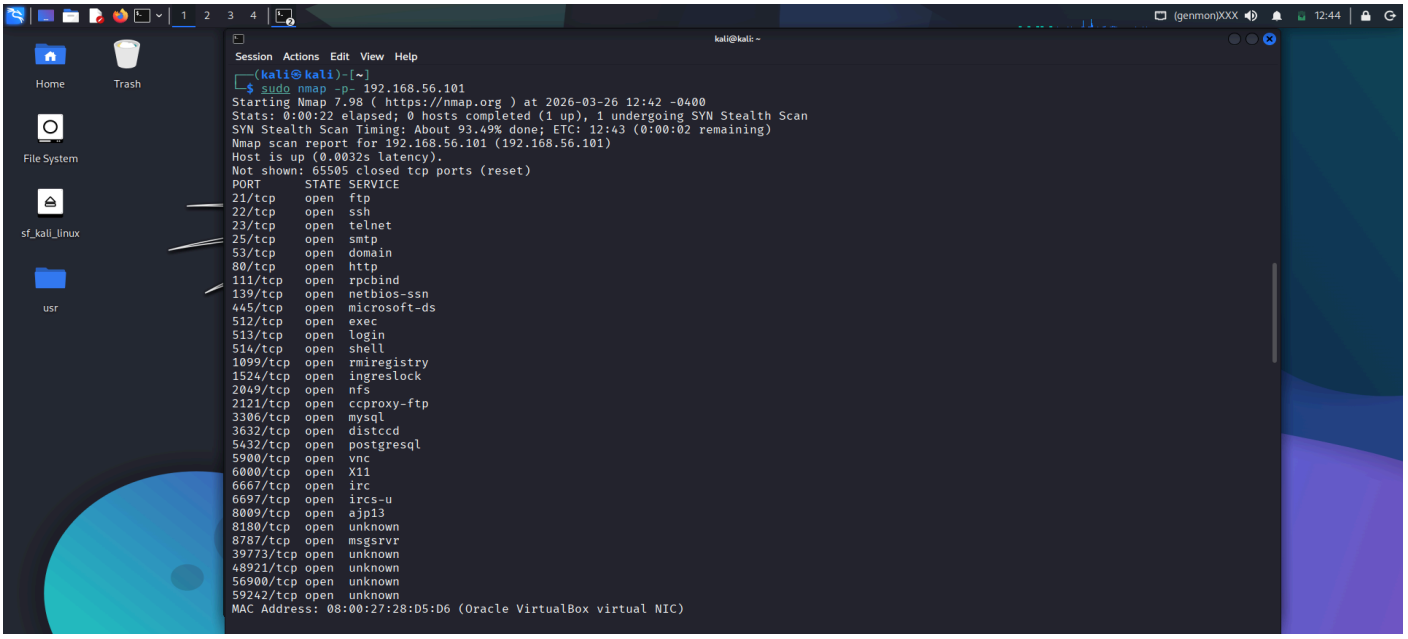

Figure 2: Nmap SYN scan showing multiple open ports on target system

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2. SSH Brute Force Attack (Ubuntu)

Hydra was used to perform SSH brute-force attack.

Command:
hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.110

Timestamp: 06:57

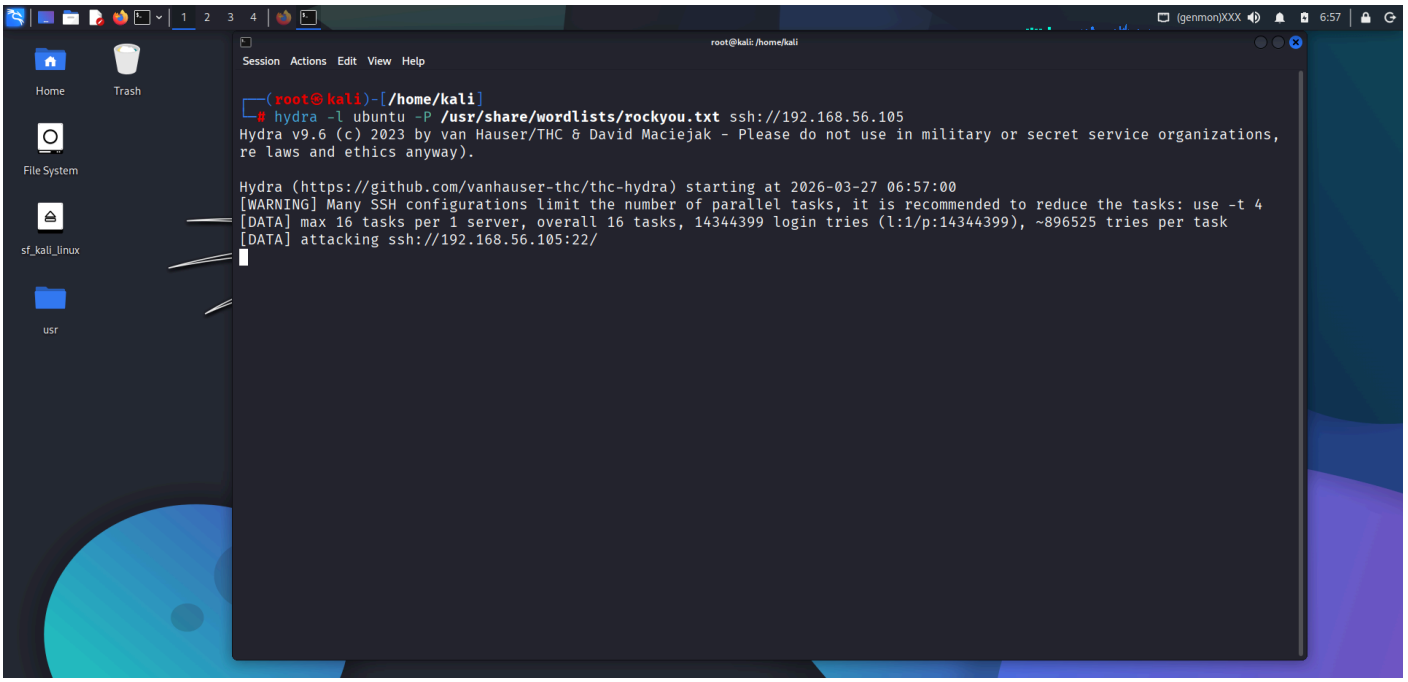

Figure 3: SSH brute force attack

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3. RDP Brute Force Attack (Windows)

Hydra was used to perform RDP brute-force attack.

Command:
hydra -t 4 -V -f -l Administrator -P /usr/share/wordlists/rockyou.txt rdp://192.168.56.105

Timestamp: 07:45

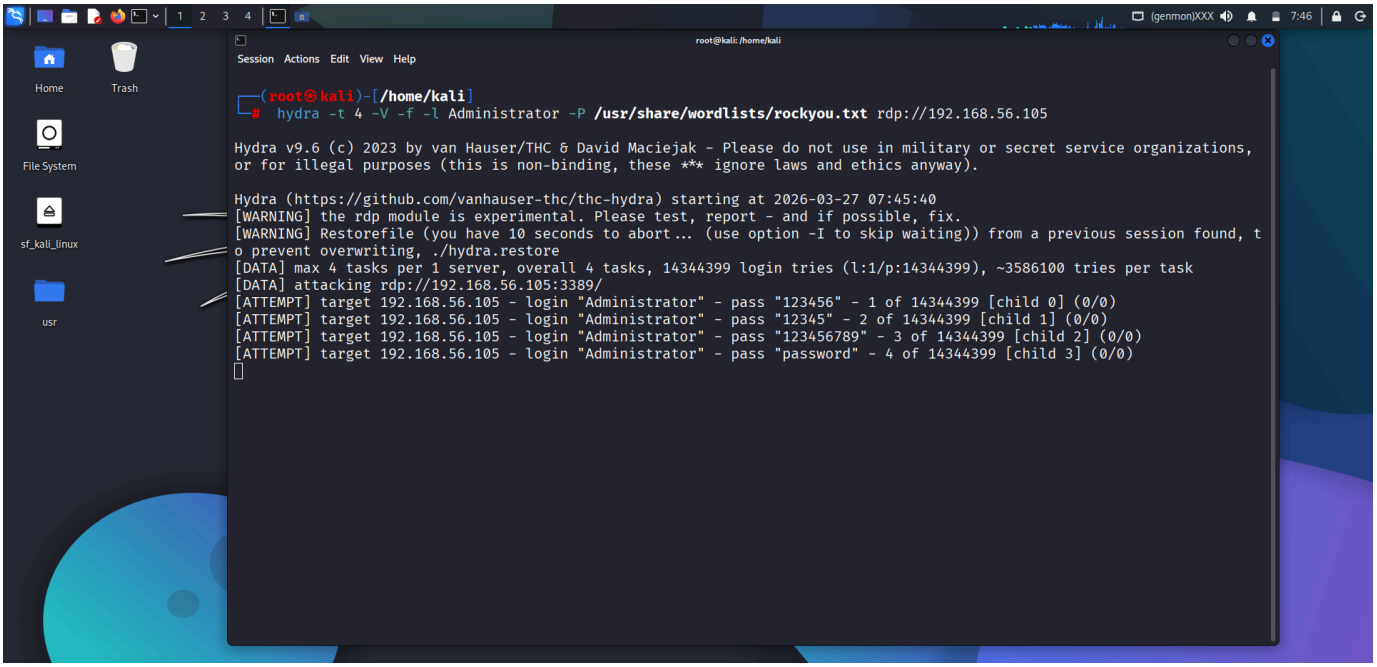

Figure 4: RDP brute force attack

Detection in Wazuh

Ubuntu Alerts

Wazuh detected authentication failures.

Timestamp: 07:00 Wazuh detected authentication failures (Ubuntu)

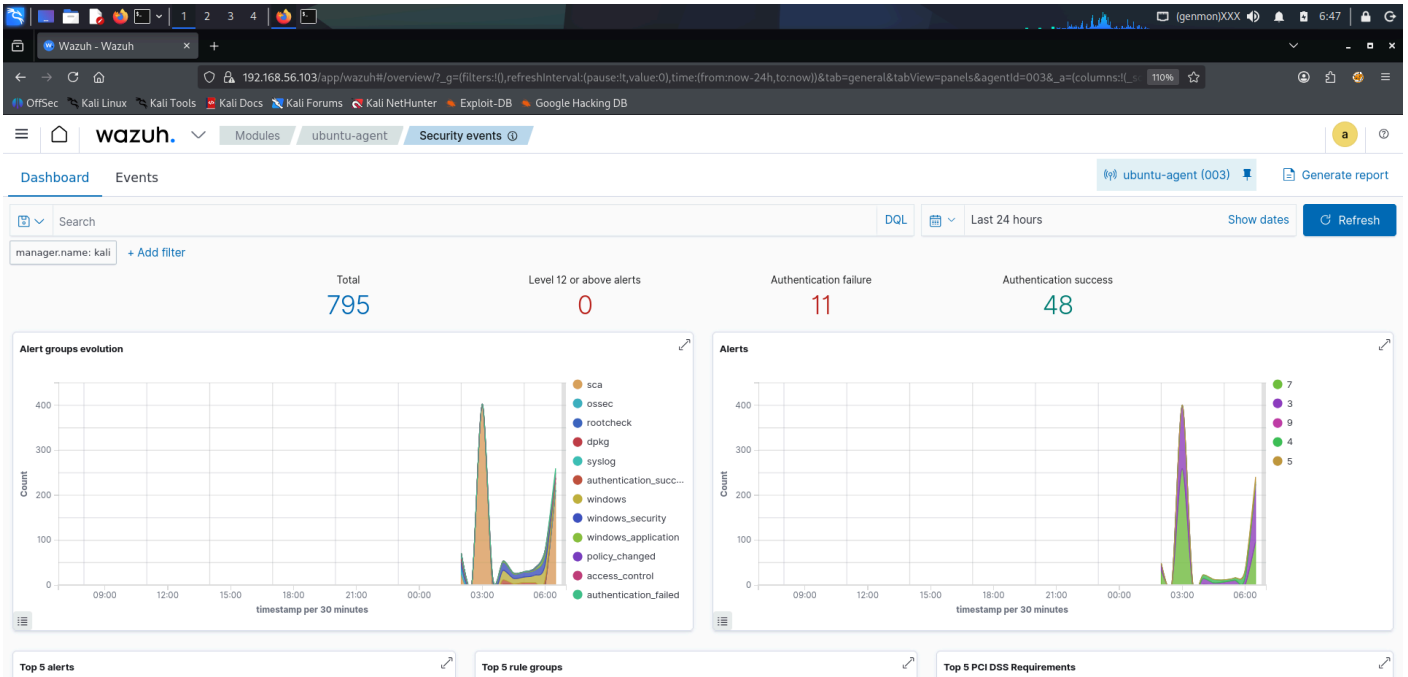

Figure 5: Ubuntu authentication alerts

Windows Alerts

Wazuh detected failed privileged operations.

Timestamp: 07:39

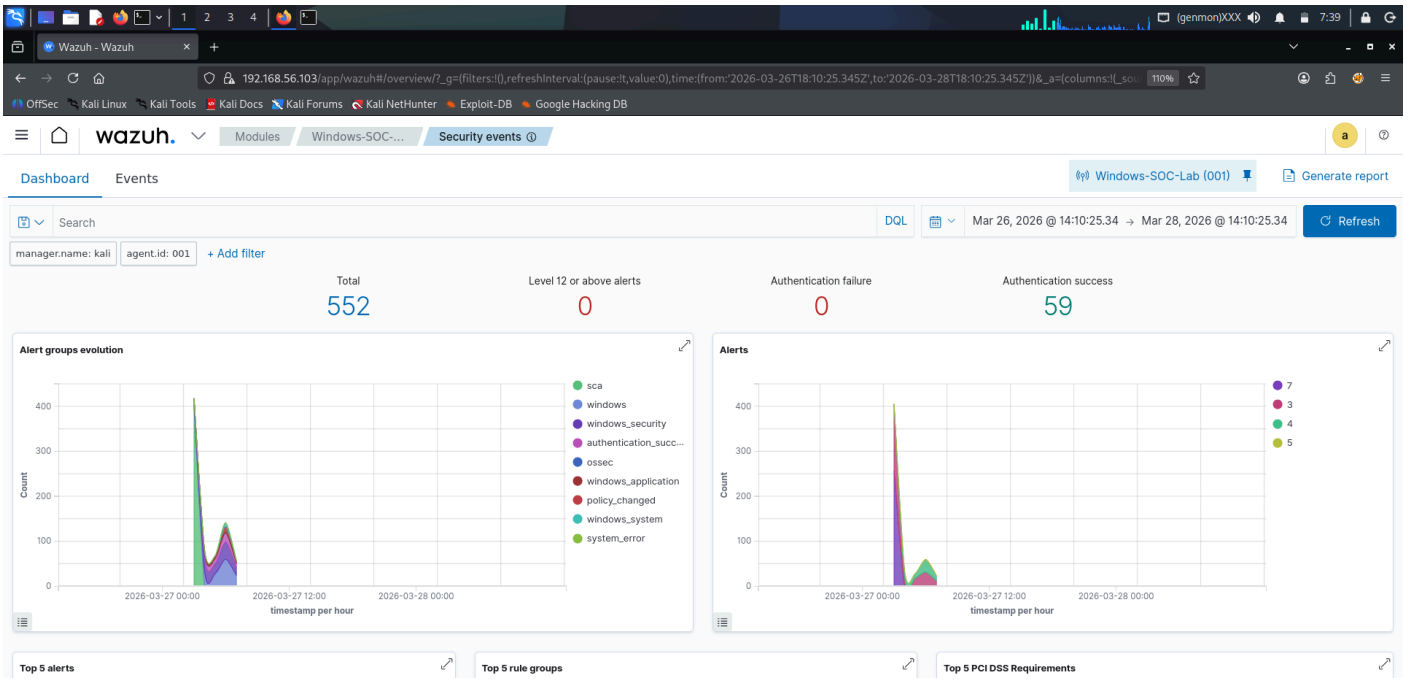

Figure 6: Windows authentication alerts

Timeline of Events

Time	Activity	Target
06:45	Nmap Scan	Ubuntu
06:57	SSH Brute Force	Ubuntu
07:00	Authentication Failures Detected	Ubuntu
07:39	RDP Brute Force	Windows
07:45	Wazuh Alert	Windows

Analysis

The SSH brute-force attack generated multiple authentication failure logs. Wazuh successfully detected these attempts and triggered alerts. The RDP brute-force attack resulted in failed privileged operation alerts on Windows. Both attacks were logged and categorized correctly.

Recommendations

- Implement account lockout policies
- Use strong passwords
- Enable firewall restrictions
- Limit SSH login attempts
- Monitor logs continuously

Conclusion

The lab demonstrated that Wazuh effectively detects brute-force and reconnaissance attacks. Both Ubuntu and Windows agents successfully generated alerts. The monitoring system provides real-time visibility and supports incident response.